

SMART WATERING SYSTEM

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ROB GY-5103

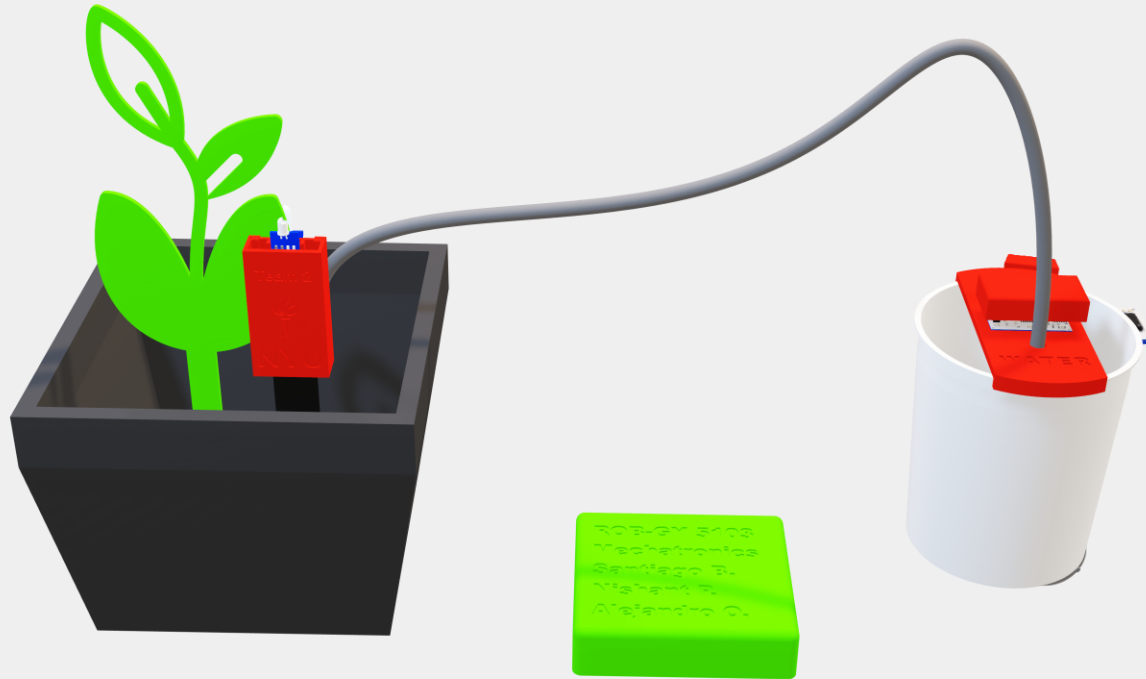


UNDERSTANDING SOIL MOISTURE

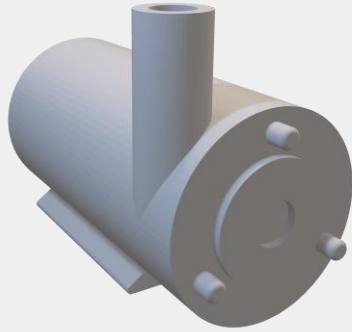
Understanding moisture level in plants in different conditions such as droughts, help mitigate the biggest thread to world food security. [1]



THE COMPLETE SYSTEM



THE WATER TANK SUBSYSTEM



WATER PUMP

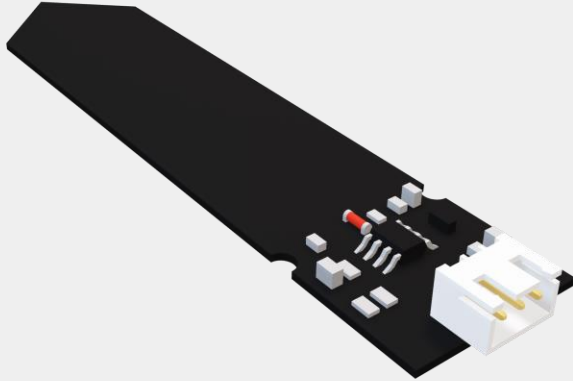


ULTRASONIC SENSOR

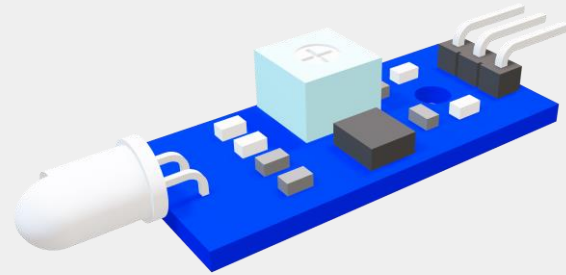


THE PLANT SUBSYSTEM

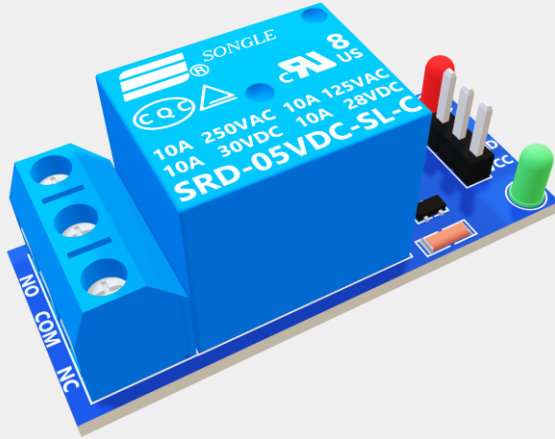
MOISTURE SENSOR



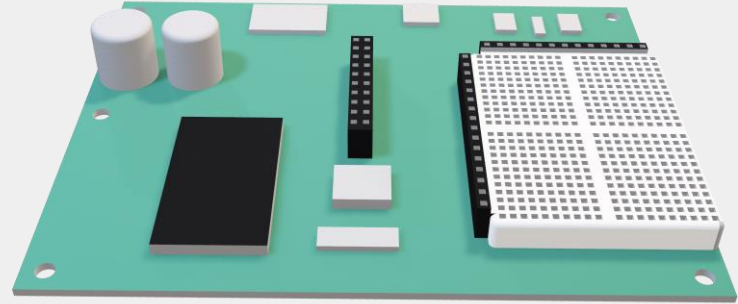
LIGHT SENSOR



THE CONTROL SUBSYSTEM



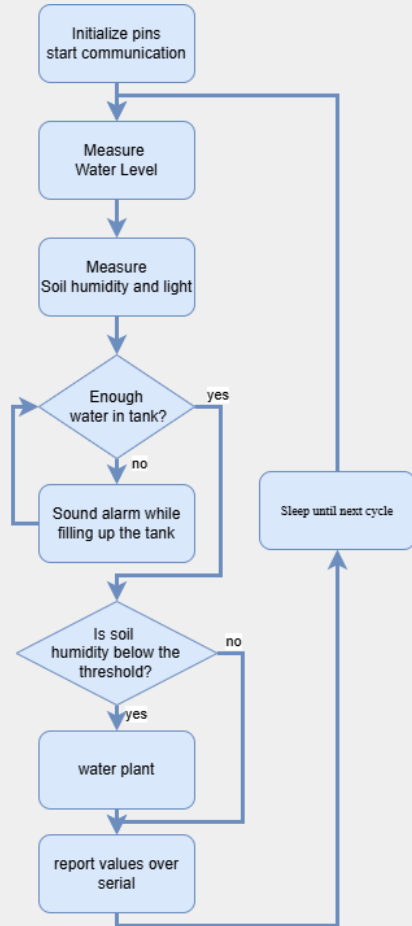
RELAY



BASIC STAMP 2



FLOW DIAGRAM

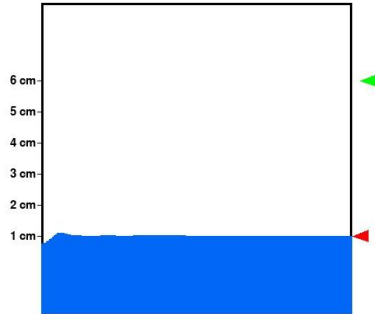


USER INTERFACE

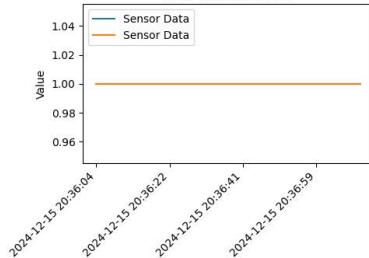
Water Level Measurement

Water Level Percentage: 0%

Stored Time: DD:MM:YYYY HH:MM:SS



Real-Time Data Plot



Refilling the tank

Last Refill On: 2024-12-15 20:37:10%



Watering the Plant

Last Watering On: 2024-12-15 20:37:10%



Light Measure



PARALLAX

SYSTEM LIMITATIONS



1

**COMPUTATIONAL
POWER**



2

**RELIABLE BACKUP
WATER SUPPLY**



3

**INDOOR
ENVIROMENT**



4

**INPUT PIN
LIMIT**

BILL OF MATERIALS

Number	Name	Qty	Price/Unit	Total Price
1	Basic Stamp 2	1	38.22	38.22
2	Education Board	1	54.59	54.59
3	Power Supply	1	14.99	14.99
4	Relay Module	2	1.40	2.80
5	Water Pump	2	2.50	5.00
6	Moisture Sensor	1	2.20	2.20
7	Ultrasonic Sensor	1	3.95	3.95
8	Light Sensor	1	0.89	0.89
9	Structure/Printing	1	6.00	6.00
Total				128.64



FUTURE WORK



**INTEGRATION WITH
IOT PLATFORMS**



**ENHANCED SENSOR
SUITE**



**MACHINE LEARNING FOR
ADAPTIVE WATERING**



**MULTI-PLANT
MANAGEMENT**



**ENHANCED USER
INTERFACE**



**DURABILITY AND
WEATHERPROOFING**

REFERENCES

[1] Seleiman MF, Al-Suhaibani N, Ali N, Akmal M, Alotaibi M, Refay Y, Dindaroglu T, Abdul-Wajid HH, Battaglia ML. Drought Stress Impacts on Plants and Different Approaches to Alleviate Its Adverse Effects. *Plants (Basel)*. 2021 Jan 28;10(2):259. doi: 10.3390/plants10020259. PMID: 33525688; PMCID: PMC7911879.

[2] M. R. H. Naeem, S. Gawhar, M. B. H. Adib, S. A. Sakib, A. Ahmed and N. A. Chisty, "An IoT Based Smart Irrigation System," 2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), DHAKA, Bangladesh, 2021, pp. 243-247, doi: 10.1109/ICREST51555.2021.9331092.



THANKS!

Do you have any questions?

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